



Lab Sheet 3

Database Management System Lab (BTCS203P)

B. TECH AI&ML, CYBER SECURITY, Data Science 3RD Semester (Session 2026-2025)

Objective:

To understand and practice SQL Functions.

Prerequisites:

1. Basic knowledge of SQL.
2. MySQL installed and set up.

Lab Sheet 2: SQL Functions

A. Number Functions

1. Demonstrate ABS() with suitable examples.

```
90 • SELECT ABS(-25) AS Absolute_Value;
```

Absolute_Value
25

1. Demonstrate ROUND() with suitable examples.

```
91 • SELECT ROUND(125.678, 2) AS Rounded_Value;
```

Rounded_Value
125.68

2. Demonstrate CEIL() and FLOOR().

```
92 • SELECT CEIL(12.34) AS Ceiling_Value,  
93 FLOOR(12.78) AS Floor_Value;
```

Ceiling_Value	Floor_Value
13	12

3. Demonstrate MOD() with suitable examples.

```
94 • SELECT MOD(17, 5) AS Remainder;  
95
```

Remainder
2

4. Calculate the rounded salary of employees.

```
95 • SELECT Name, Salary, ROUND(Salary, -2) AS Rounded_Salary FROM Employee;
```

Name	Salary	Rounded_Salary
Amit	42350.00	42400
Rahul	50820.00	50800
Neha	60500.00	60500
Priya	45980.00	46000
Rohit	33880.00	33900

B. Aggregate Functions

6. Find the total number of students using COUNT().

```
96 • SELECT COUNT(*) AS Total_Students FROM Student;
```

Total_Students
14

7. Find the total salary of all employees using SUM().

```
97 • SELECT SUM(Salary) AS Total_Salary FROM Employee;
```

Total_Salary
233530.00

8. Find the average salary using AVG().

```
98 • SELECT AVG(Salary) AS Average_Salary FROM Employee;
```

Average_Salary
46706.000000

9. Find the highest salary using MAX().

```
99 • SELECT MAX(Salary) AS Highest_Salary FROM Employee;
```

Highest_Salary
60500.00

10. Find the lowest salary using MIN().

```
100 • SELECT MIN(Salary) AS Lowest_Salary FROM Employee;
```

Lowest_Salary
33880.00

C. Character Functions

11. Demonstrate UPPER() on student names.

```
101 • SELECT Name, UPPER(Name) AS Upper_Name FROM Student;
```

Name	Upper_Name
Anshika	ANSHIKA
Radha	RADHA
Jayshree	JAYSHREE
Ruhani	RUHANI
Shreya	SHREYA
Maya	MAYA
Rahul	RAHUL
Priya	PRIYA
Aman	AMAN
Neha	NEHA
Rohit	ROHIT
Simran	SIMRAN

12. Demonstrate LOWER() on student names.

```
102 • SELECT Name, LOWER(Name) AS Lower_Name FROM Student;
```

Name	Lower_Name
Anshika	anshika
Radha	radha
Jayshree	jayshree
Ruhani	ruhani
Shreya	shreya
Maya	maya
Rahul	rahul
Priya	priya
Aman	aman
Neha	neha
Rohit	rohit

13. Demonstrate LENGTH() on student names.

```
103 • SELECT Name, LENGTH(Name) AS Name_Length FROM Student;
```

Name	Name_Length
Anshika	7
Radha	7
Jayshree	8
Ruhani	6
Shreya	6
Maya	4
Rahul	5
Priya	5
Aman	4
Neha	4
Rohit	5
Simran	6

14. Demonstrate CONCAT() using first and last names.

15. Demonstrate SUBSTRING()/SUBSTR() with suitable examples.

```
104 • SELECT First_Name, Last_Name, CONCAT(First_Name, ' ', Last_Name) AS Full_Name FROM Student;
105 • SELECT Name, SUBSTRING(Name, 1, 3) AS First_Three_Characters FROM Student;
```

Name	First_Three_Characters
Anshika	Ans
Radha	Rad
Jayshree	Jay
Ruhani	Ruh
Shreya	Shr
Maya	May
Rahul	Rah
Priya	Pri
Aman	Ama
Neha	Neh
Rohit	Roh
Simran	Sim

D. Conversion Functions

16. Convert a number into a character/string value.

```
106 • SELECT CAST(12345 AS CHAR) AS String_Value;
```

String_Value
12345

17. Convert a string into a numeric value.

```
107 • SELECT CAST('12345' AS UNSIGNED) AS Numeric_Value;
```

Numeric_Value
12345

18. Convert a date value into a different format.

```
108 • SELECT DATE_FORMAT('2026-09-17', '%d-%m-%Y') AS Formatted_Date;
```

Formatted_Date
17-09-2026

19. Demonstrate CAST() with suitable examples.

```
109 • SELECT CAST(125.75 AS DECIMAL(10,2)) AS Converted_Value;
```

Converted_Value
125.75

20. Demonstrate CONVERT() where supported by the DBMS.

```
110 • SELECT CONVERT(12345, CHAR) AS String_Value;
```

String_Value
12345

E. Date Functions

21. Display the current date using a suitable date function.

```
111 • SELECT CURDATE();
```

CURDATE()
2026-09-17

22. Extract the year, month and day from a date.

```
112 • SELECT YEAR('2026-09-17') AS Year, MONTH('2026-09-17') AS Month, DAY('2026-09-17') AS Day;
```

Year	Month	Day
2026	9	17

23. Find the difference between two dates.

```
113 • SELECT DATEDIFF('2026-09-17', '2026-01-01') AS Date_Difference;
```

Date_Difference
259

24. Add a specified number of days to a date.

```
114 • SELECT DATE_ADD('2026-09-17', INTERVAL 10 DAY) AS New_Date;
```

New_Date
2026-09-27

25. Display employees' joining dates in a required date format.

```
.15 • SELECT Name, Joining_Date, DATE_FORMAT(Joining_Date, '%d-%m-%Y') AS Formatted_Joining_Date FROM Employee;
```

Note: Function names can vary slightly between MySQL, Oracle and SQL Server. Since you use MySQL Workbench, these can be framed specifically for MySQL.

Submission:

1. Take screenshots of each step's output.
2. Save the SQL scripts in a text file.